BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company of New York, Inc. for Electric and Gas Service.

Cases 25-E-0072, 25-G-0073

June 24, 2025

Rebuttal Testimony of:

Depreciation Panel

Qun Li
4 Irving Place
New York, New York 10003

Robert Benoit
4 Irving Place
New York, New York 10003

Melissa M. Howard
207 Senate Avenue
Camp Hill, Pennsylvania

Ned W. Allis
207 Senate Avenue
Camp Hill, Pennsylvania

Case Nos. 25-E-0072 and 25-G-0073

CONSOLIDATED EDISON COMPANY OF NEW YORK, INC.

DEPRECIATION PANEL REBUTTAL TESTIMONY

TABLE OF CONTENTS

I. REBUTTAL 2

A. AVERAGE SERVICE LIFE AND SURVIVOR CURVE ESTIMATES 9

1. Gas Service Lives and the Impact of the CLCPA 15

i. Introduction 15

ii. The CLCPA Proceeding Demonstrates the Short- and Long-Term Tradeoffs for Depreciation 18

iii. The Commission Does Not Need to Wait for Further Analysis to Adopt the Company's Proposal 22

iv. The Company's Proposal is Moderate When Compared to CLCPA Scenarios 24

v. Depreciation Should Recognize the Impacts of the CLCPA 28

vi. The CLCPA and Obsolescence 36

vii. Response to Mr. Garrett's Arguments..... 41

viii.Conclusion..... 45

2. Electric Service Life Estimates 46

B. NET SALVAGE 50

C. RESERVE VARIATIONS 53

II. CONCLUSION 53

Q. Would the members of the Depreciation Panel ("Panel") please state their names.

A. Qun Li, Robert Benoit, Ned W. Allis, and Melissa M. Howard.

Q. Have you previously testified in this proceeding?

A. Yes. We previously submitted direct testimony as the Panel on behalf of Consolidated Edison Company of New York, Inc. ("Con Edison", "CECONY", or the "Company"). We provided our credentials as part of our direct testimony.

Q. What is the purpose of your rebuttal testimony?

A. The purpose of our rebuttal testimony is to respond to the depreciation-related proposals and discussions included in the direct testimonies of the Staff Depreciation Panel ("SDP" or "Staff"), the Utility Intervention Unit ("UIU") witness David Garrett, New York City ("NYC") witness John Sano, and New York Energy Consumers Council, Inc. ("NYECC") witness Andy Anderson.

I. REBUTTAL

Q. Please summarize your rebuttal testimony.

A. Our rebuttal testimony responds to the depreciation-related proposals made by Staff, UIU, NYC, and NYECC. Most of these proposals are in relation to the recommended service lives in this proceeding, as well as, for gas assets, the

related issue of how or whether the impacts of the Climate Leadership and Community Protection Act ("CLCPA") should be incorporated into the recommended service lives and depreciation rates. A brief summary of each proposal is as follows:

- Staff proposes changes to the service life estimates for several electric accounts and several gas accounts (Exhibit_(SDP-3)). Staff also proposes different net salvage estimates for several gas and electric accounts (Exhibit_(SDP-4)). For the gas accounts, Staff recommends that any recognition of the CLCPA be deferred until the Commission has analyzed the results of depreciation analyses directed in the Commission's May 12, 2022 Order ("Gas Planning Order") in Case No. 20-G-0131 ("Gas Planning Proceeding"). However, as we discuss, the depreciation analysis in that proceeding, provided more than two years ago, demonstrated convincingly that depreciation will need to increase and, moreover, it will likely need to increase more than the Company's proposal in this case. There is ample support to reduce service lives, as we have

recommended, and delaying action further will harm future customers who will have to bear a disproportionate share of the energy transition.

- UIU proposes changes to the service life estimates for several electric and gas accounts (Exhibit_(DJG-3)). UIU's proposals for gas assets are also based, at least in part, on an argument that the Company should defer recognizing the impact of the CLCPA.
- NYC argues (at 10) that we should defer the recognition of the CLCPA on depreciation, although NYC does not make specific proposals for service lives or net salvage.
- NYECC proposes (at 21) rejecting the Company's reductions in service lives based on the "excessive" revenue requirements that result. NYECC also incorrectly alleges (at 17) that the Company is more concerned about the cost impact to future ratepayers than to current ratepayers.

To the extent that an issue is raised by Staff or intervenors but not addressed specifically by this Panel,

the positions stated therein should not be viewed or considered to have been accepted by the Company.

Q. Do you agree with the referenced parties' proposals?

A. No. There are three primary areas of disagreement. First, all four parties disagree with taking any action to address the cost recovery implications of the CLCPA. Second, Staff and UIU propose longer service lives for certain electric accounts. Finally, Staff recommends less negative net salvage estimates for several gas and electric accounts. Each of these changes also impact the theoretical reserve imbalance, although the approach recommended by each party – to amortize the portion above 10 percent of the theoretical reserve – is consistent.

We disagree with each of these recommendations. The Company's current service lives are long by industry standards, and the Company's net salvage estimates are less negative than supported by the historical data in most instances. The Company has not sufficiently recovered its costs through depreciation due to its previous depreciation rates being too low. Staff's and UIU's proposals would continue this trend. Further, the failure of all four parties to incorporate the impacts of the CLCPA will result

in significant under-recovery challenges in the future and will be harmful to future customers.

Q. Please explain further the issues with the parties' proposals related to the impacts of the CLCPA.

A. With regard to the impacts of the CLCPA, each party effectively proposes that we should not incorporate any potential impacts of the CLCPA on depreciation and instead suggest that we should defer any consideration to a future proceeding. The reasons put forward for doing so vary to some degree, but each party suggests that it would be more appropriate to delay taking any action until more information is available – either until there is further direction from the Commission in the Gas Planning Proceeding, or, more generally, until there is more certainty about the specific impacts of the CLCPA on the gas business. Further, the parties have incorrectly implied that our recommendation favors future customers at the expense of current customers.

We believe these suggestions are misguided for several reasons. First, there is sufficient information available today to, at a minimum, adopt the Company's proposal. We can be reasonably certain that the CLCPA's goals will have

an impact on the gas system and that it is highly likely that depreciation will need to increase due to a combination of shorter service lives and declining consumption. Under Case 22-M-0149, the Commission stated in its May 12, 2022 Order ("CLCPA Order") that "achievement of the CLCPA's GHG emissions mandates will require significant reductions in the use of natural gas" (at 25). It logically follows that depreciation needs to be higher for gas assets than it is today – and perhaps significantly higher. As we have shown in our direct testimony and will demonstrate further in this rebuttal testimony, the Company's proposal is very conservative when compared to potential gas planning scenarios.

Second, the Company has already conducted several analyses of the potential impacts of the CLCPA on depreciation. This includes the Gas Planning Depreciation Study, a special depreciation study performed for Con Edison by Concentric Energy Advisors ("Concentric Study" in our testimony) filed in May 2021 in Case No. 19-G-0066 and in the scenarios presented in our direct testimony. Generally, these analyses result in higher – and often much higher – depreciation than the Company has proposed in this

case. The various analyses Con Edison has already performed provide a clear demonstration that the Company's proposal is quite conservative and, as we discussed in our direct testimony, the minimum movement in depreciable lives that is reasonable in the context of the CLCPA.

Third, and perhaps most important, it is both costly and inequitable to wait to recognize the impacts of the CLCPA. The target date for reducing greenhouse gas emissions of 2050 is only 25 years from now. Con Edison is already aggressively implementing its clean heat program to convert customers to electrified heating. The longer the Commission waits to recognize the potentially significant impacts of the CLCPA, the fewer years that will remain to recover costs. Waiting to incorporate the impacts of the CLCPA on depreciation means that depreciation will have to increase even more in future years and that rate base will be higher. Both will cause even sharper increases in cost to customers and, if fewer customers remain at that point in the future, the increases for each customer will be even more pronounced and inequitable. To be clear, the sooner the Commission begins to recognize the costs of the CLCPA

in depreciation, the less costly the inevitable transition will be.

A. AVERAGE SERVICE LIFE AND SURVIVOR CURVE ESTIMATES

Q. Please summarize the average service life and survivor curve proposals of each party.

A. There are several parties that discuss the proposed service life estimates. These recommendations can be divided into two broad categories. The first is that both Staff and UIU have made specific service life recommendations for several accounts that differ from those proposed by the Company. These affect electric and gas accounts. For electric accounts, the differences are based on interpretations of the data and other relevant factors – that is, these differences are the type one would find in a typical, "business-as-usual" depreciation study.

The second is related to the Company's recommendations based on the impact of the CLCPA on gas depreciation. Staff, UIU, NYC, and NYECC all differ from the Company on these recommendations and each proposes to, effectively if not explicitly, ignore any potential impacts of the CLCPA

at this time and instead recommend service life estimates that would be more reasonable if the CLCPA did not exist.

Q. How will you address the various proposals related to service lives?

A. Given that there are different considerations and rationales underlying the recommendations of the Company and certain parties for gas and electric assets, we will address each type of service separately. We first discuss gas service lives and the CLCPA, as this is the most significant issue from a policy standpoint and is addressed by more parties. We will then discuss the more specific recommendations for electric assets. However, we should be clear that, while we expect the CLCPA to have the most significant impact on gas depreciation, its potential effects on the electric industry will have to be taken into account at some point, although we have no specific recommendations for shorter service lives for electric assets in this rate filing.

Q. Please summarize the proposals for gas assets.

A. Staff and UIU have each proposed average service life and survivor curve estimates for several gas accounts that differ from our recommendations. NYC and NYECC do not

propose specific estimates for any accounts and instead propose to maintain the current estimates. Most of the differences between our proposals for gas assets, the depreciation rates used in the Company's revenue requirement, and the proposals of other parties result from different judgements on how the impact of the CLCPA should be incorporated into depreciation. As we will explain, this would be inequitable and unfair to future customers as we should reasonably expect that the CLCPA will result in the future service life experiences to be different from those in the past. For electric assets, the primary differences are judgements of interpretation of the historical data and other factors, which are detailed in this testimony.

Q. Please provide a comparison of each proposal for gas assets.

A. Table 1 of Exhibit___(DP-7) provides a comparison of the service life proposals of each party for gas assets. Included in the table are the current average service life estimates as well as the "Base Case" scenario, which represents the average service lives that are, in our judgement, appropriate under "business-as-usual" conditions

if the CLCPA did not exist. We have also shown the "5-Year Shorter" scenario, which is our proposal, and the "10-Year Shorter" scenario. The table also shows Staff's and UIU's recommendations. While there are some exceptions, for many of these accounts, their proposals are similar to the Base Case scenario, which is generally consistent with their recommendations to defer recognizing the impact of the CLCPA until a later time. We note that because NYC and NYECC have not made specific proposals, we do not show specific lives for their proposals. However, their proposals are, in effect, either the Current or Base Case scenarios.

Q. Please summarize the proposals for electric assets.

A. The only parties that recommend changes to the service lives for electric assets are Staff and UIU. Staff proposes changes for two accounts. UIU proposes a different estimate for one of these same accounts (Account 367), as well as two additional electric accounts. Table 2 of Exhibit___(DP-7) provides a summary of the current estimates, our recommendations and the recommendations of Staff and UIU for these four accounts.

Q. Do you have any other general observations on the service life proposals?

A. Yes. As we discussed in our direct testimony, the current service lives for Con Edison for many accounts are relatively long when compared to the industry. Indeed, as can be seen in Exhibit___(DP-7), even with the 5-Year reduction in service lives we recommend, the recommended average service lives for the largest gas accounts, such as mains and services, are still within the typical industry range and are closer to the middle or upper end of the range than the lower end of the range. Accordingly, our recommendations incorporate an expectation that most of the gas assets will continue to provide service beyond 2050. Additionally, for most of the electric accounts at issue, our recommended lives are at or beyond the upper end of the industry range, and the proposals of Staff and UIU are mostly beyond the upper end of the range for these accounts.

Q. Why do you believe this is important?

A. We believe this perspective is important as the Commission considers the various depreciation proposals in this case, because the CLCPA will result in a profound transformation

of the way in which energy is delivered to New York consumers. Indeed, whether through the scenarios outlined in the draft scoping memo of the NY Climate Action Council or the various scenarios requested by the Commission in the Gas Planning Proceeding, many parties, as well as the Commission in its CLCPA Order, expect significant impacts on gas transmission and distribution systems. These impacts are likely to mean that it will be equitable and necessary to have higher depreciation expense for Con Edison than is the case today and that service lives are likely to be shorter for many assets than the historical experience of Con Edison.

With this in mind, the Company's proposal is realistically only a modest recognition of the potential impacts of the CLCPA. While we accept that there is uncertainty and a range of future scenarios, based on our experience most, if not all, of these scenarios would result in higher depreciation than the Company's proposal. As the Commission stated, it expects a significant reduction in gas consumption. The longer the Commission waits to begin incorporating these impacts, the more costly

it will be to customers and the higher the risk that the energy transition envisioned by the CLCPA will not succeed.

1. Gas Service Lives and the Impact of the CLCPA

i. Introduction

Q. Please explain the differences between the service life proposals for gas assets.

A. The primary difference is that the Company's recommendations incorporate the impacts that a moderate (or hybrid) decarbonization scenario would have on the service lives of the Company's assets. The other parties providing testimony on the issue - Staff, UIU, NYC and NYECC - do not believe the impacts of the CLCPA on gas depreciation should be incorporated into the Company's revenue requirement. In Exhibit___(DP-4), we provided several different depreciation scenarios. We have updated these analyses to include Staff's and UIU's proposals in Exhibit___(DP-8). These analyses show that the Company's proposal is conservative when compared to most CLCPA scenarios and that the proposals of the other parties do not sufficiently increase depreciation to account for CLCPA impacts. As we discuss later, if depreciation is too low today, not only

will future customers be disproportionately harmed, the total costs of the gas transition increase.

Q. In your judgement, would there be any reason to expect depreciation to be lower than your proposal?

A. No. First, the impacts of the CLCPA, or of decarbonizing by 2050 in general, have been extensively studied. The Commission has had an open docket, under Case 20-G-0131, on the issue since March 2020 and the Company, as well as the other large New York Local Distribution Companies ("LDCs"), filed studies assessing the potential impact on depreciation in November 2022. The Company's analysis demonstrated that under the scenarios considered by the Commission, depreciation would need to increase. Further, the Company's proposal in the instant case, which is to slightly shorten service lives of gas assets, is perhaps the most moderate step the Commission could take at this point in time. As we discuss below, further investigation in other jurisdictions has revealed similar results.

Based on these analyses, the only way for depreciation to be lower than the Company's depreciation scenario, as proposed by the other parties, is if the state were to abandon the CLCPA's goals. However, depreciation would

still be too low under the business-as-usual scenario. This is because business-as-usual in New York has been for utilities to have long depreciable lives and net salvage estimates that are not sufficiently negative when compared to the historical data.

The Company's depreciable service lives, and those of other New York utilities, are longer than most in the industry. In our experience, one reason for this is that longer service lives, and less negative net salvage, have typically been adopted in rate cases (usually through settlement). For example, the current average service life "ASL" estimate for gas mains is 80 years. The Company's proposal to shorten service lives still results in an ASL of 75 years. This is still a long service life for this account by industry standards. In a scenario in which the gas system does not attain the CLCPA goals, we would still expect pipe to be replaced at a faster rate than has occurred historically.

In summary, we do not see any scenario in which the Company's depreciation proposal for gas assets would be too high. It is more likely that depreciation will need to be higher in the future and potentially significantly higher

than the Company's proposal. We are quite certain that the depreciation proposals of Staff, UIU, NYC and NYECC are too low for adequate recovery.

ii. ***The CLCPA Proceeding Demonstrates the Short- and Long-Term Tradeoffs for Depreciation***

Q. Certain parties claim that the Company's proposal is based on a lack of consideration for current customers or a desire to recover costs more quickly or benefit shareholders, i.e., page 21 of NYECC witness Anderson's testimony and page 42 of NYC witness Sano's testimony. Please address these criticisms.

A. The Company's depreciation proposal is based on the recommendations of Mr. Allis and Ms. Howard, outside depreciation experts who are not employed by the Company. These recommendations are consistent with their proposals both in prior cases for the Company and in other cases and jurisdictions and incorporate work they have performed on both the short- and long-term implications of the CLCPA and decarbonization in general. The Company's recommendations consider the implications to both current customers and future customers, as well as how the customer base will change both in terms of composition and gas usage per

customer. Contrary to the implication that our proposal favors future customers at the expense of current customers, it is more likely that our recommendations are for too little, not too much, depreciation. That is, if our proposal is more favorable to one generation of customers, it is current customers rather than future customers.

Q. Please explain.

A. In our analysis provided in the November 2022 Depreciation Study in Case No. 20-G-0131 ("CLCPA Depreciation Study"), we analyzed three scenarios related to the future of the gas industry: 1) a Reference Scenario; a 2) Hybrid Scenario; and a 3) Deep Electrification Scenario. Generally, throughput and the number of gas customers declined most significantly in the Deep Electrification Scenario, with smaller declines in the hybrid scenario (although consumption declines more significantly on a per-customer basis than the Reference Scenario). The projections for these scenarios were subsequently updated in the Company's Long-Term Gas Plan.

To illustrate the short- and long-term impacts of depreciation proposals, we have provided results from

similar analysis performed in the Case No. 20-G-0131 depreciation studies with the projections from the Company's LTGP in Exhibit___(DP-9). The overall results are similar to the conclusions from our original analysis, which underscores our expectation that there is not a contemplated scenario that meets CLCPA goals that would not require higher depreciation. Figure 1 of Exhibit___(DP-9) provides, as a proxy for the bill impact to customers, the estimated future revenue requirement per customer on an annual basis, assuming the Commission does not change the approach to depreciation and depreciation is not increased to account for shortening service lives and declining consumption. As the chart illustrates, particularly for a deep electrification scenario, future customers will bear a much higher share of revenue requirements if there are no changes to depreciation.

However, this impact can be mitigated with higher depreciation expense today. Figures 2 and 3 of Exhibit___(DP-9) show the long-term impacts of several approaches, including the Company's proposal to shorten service lives as well as the Units of Production method, for each of these scenarios. This analysis first shows

that the hybrid scenario is significantly less costly to customers in the long-term than the deep electrification scenario. Further, for the deep electrification scenario, depreciation would have to be significantly higher than the Company's proposal. The Units of Production method would be needed for this scenario to mitigate future cost increases. It should be noted that, in our analysis, even the Units of Production method did not fully bend the curve to mitigate future cost increases. As Mr. Allis has demonstrated more recently, combining the UoP Method with both shorter service lives and the ELG procedure is necessary to truly achieve equitable depreciation with significant declines in demand.

The hybrid scenario would allow for more equitable depreciation with lower costs to current customers than the phased electrification scenario. However, even here depreciation would need to be higher than the Company's proposal to achieve intergenerational equity.

Q. What can you conclude from these analyses?

A. A primary conclusion is that depreciation will need to increase and be higher than the Company's proposal in this case, to achieve an equitable sharing of costs over the

long-term. This conclusion has been further supported in more recent work by Gannett Fleming in Massachusetts's gas planning proceeding, DPU 20-80, e.g., Exhibit NG-CCP, Att 6 in Massachusetts DPU 25-41, filed April 1, 2025.

As a result, the other parties' criticisms are unwarranted. We are unable to identify a realistic scenario in which New York achieves CLCPA goals and depreciation does not need to increase beyond our proposal in this case for an equitable sharing of costs.

iii. The Commission Does Not Need to Wait for Further Analysis to Adopt the Company's Proposal

Q. Staff, UIU, NYC, and NYECC each argue that there needs to be guidance from the Commission in Case 20-G-0131 before adopting any depreciation impacts from the CLCPA. Do you agree?

A. No. First, the mere existence of a gas planning proceeding does not preclude the Commission from adopting more appropriate depreciation rates in a utility rate case. Indeed, given the long-term impacts as well as the harm to customers in delaying more equitable depreciation, there is a far stronger argument that the Commission should act as soon as practicable. Second, there is ample analysis and

evidence that depreciation needs to be higher under any reasonably anticipated future CLCPA scenario--the analysis demonstrating the need for higher depreciation has been available to the Commission and other parties for at least the last three years. Third, the harm to future customers will increase the longer the Commission waits. Case No 20-G-031 has been open for five years, and the depreciation analysis from that docket has been available since 2022. Finally, the Company's proposal represents a gradual move towards a more equitable level of depreciation.

Q. Parties raise the issues of the impacts of higher depreciation to current customers. Please address the concerns they raise.

A. We are mindful that higher depreciation can have a short-term impact on customer bills and that there are concerns with the cost of energy. However, we believe it is short-sighted to focus only on the short-term impact to current customers. As Exhibit___(DP-9) demonstrates, this will cause far more significant affordability challenges in the future.

Moreover, to the extent there are concerns with current customer bills, this is due in part to past

depreciation that was insufficient to equitably recover costs. That is, depreciation has been too low in the past, which results in both a higher rate base (because accumulated depreciation reduces rate base) and in higher depreciation resulting from a larger reserve deficiency. As discussed previously, the current depreciable lives for New York utilities are longer than most in the industry. There are costs to having depreciation rates that are too low. The affordability concerns raised by the other parties are largely the result of depreciation that has been too low in the past, which is in part due to proposals made by the parties in prior proceedings. The Commission should not continue to prioritize the short-term, as the other parties suggest. Doing so would only exacerbate future customer bill challenges and risk significantly more pronounced affordability challenges in the future.

iv. The Company's Proposal is Moderate When Compared to CLCPA Scenarios

Q. Does the Company's proposal incorporate an expectation that the gas system will no longer be used in the future or will be significantly smaller?

A. No. The Company's proposal produces results consistent with an expectation of only a very moderate decline in demand. Indeed, the depreciation rates we have recommended are considerably more moderate than the Commission indicated it expects in the CLCPA Order (at 25):

> …achievement of the CLCPA's GHG emissions mandates will require significant reductions in the use of natural gas.

NYC witness Sano (at 28) notes the restrictions his client has put on the use of gas in new buildings after 2027. We note that in comments in Case No. 20-G-0131, NYC observed that Local Law 97 (at 2) "is landmark decarbonization legislation that sets greenhouse gas emissions caps on large buildings and ratchets those caps down over time, further driving efficiency and electrification investments." NYC further commented (at 11) that:

> …the City requests that the Commission require the LDCs go beyond identifying NPA opportunities for LPP and propose opportunities to strategically retire gas assets in their long-term plans in a way that allows for a managed and cost-effective reduction in gas use. The LDCs would identify areas of their remaining gas system where it is practical to retire assets first. As gas demand declines due to energy efficiency and electrification, this early work would facilitate a broader, managed asset retirement program in the future.

Q. How do the referenced parties' proposals compare to the scenarios you have analyzed?

A. As described beginning on page 44 our direct testimony, we provided in Exhibit___(DP-4) the results of several different scenarios. We have updated these analyses to include Staff and UIU's proposals in Exhibit___(DP-8). As the table and figure in this exhibit show, Staff and UIU propose similar depreciation to the "business-as-usual" Base Case. NYC and NYECC propose to continue to use the Current Depreciation.

Q. Overall, what can you observe from Exhibit ___(DP-8)?

A. We first note that our proposal and the Company's revenue requirement proposal result in lower depreciation expense than any of the Units of Production scenarios (which assume that the gas system is still operating in 2050), the ELG scenario or the Recover by 2050 scenario. This helps to illustrate that the Company has made a very moderate proposal in this case and that its proposal is not "excessive."

Second, if one considers the range from the lowest scenario to the highest to be a potential distribution of possibilities – with just over $393 million on the low end

for Current Depreciation to more than $1.29 billion on the high end for the Deep Electrification Units of Production scenario - then all of the proposals in the instant case are relatively close to the lower end of this range. Staff's proposal is close to the Base Case and the other parties' proposals are close to or the same as the current depreciation rates. Even the Company's proposal is within $36 million of the Base Case scenario (i.e., the lower end of the range) and much closer to this low end than to the Units of Production or ELG scenarios. Given that we (and the Commission) find it highly unlikely that there will be no impact on the gas industry from the CLCPA that would not affect depreciation – meaning that the Base Case scenario is very unlikely – our recommendation is, in reality, a very moderate proposal.

We believe this perspective is important in showing the Company's request is not "excessive" nor is it premature in the context of the potential impacts of the CLCPA. As we discuss in the next section, depreciation should incorporate the impacts of the CLCPA. The Company's proposal is a very moderate recognition of the realities of what significant greenhouse gas reductions may do to the

service lives and equitable allocation of costs of gas assets. There is no reason to refuse to take that into account now, even if the Commission has ordered additional studies.

v. <u>Depreciation Should Recognize the Impacts of the CLCPA</u>

Q. Other parties have discussed depreciation and its impacts on customers. Please explain further how depreciation impacts a utility company and its customers.

A. Depreciation is both an important and complicated concept. It affects utility accounting, and as a result both customer rates and earnings, in more than one way. Often in rate cases, witnesses will primarily focus on only one aspect of depreciation, which is that depreciation is an expense incorporated into a utility's revenue requirement. However, it is also an expense on the company's books and, as an expense, annual depreciation accruals are a reduction to a company's earnings. Depreciation, therefore, has an effect on both revenues and expenses. This is consistent with the concept of the "matching principle" used in accounting, which provides a requirement to recognize expenses on the income statement in the same period in

which the related revenues are earned. *Depreciation Systems* by Frank K. Wolf and W. Chester Fitch, a widely cited depreciation text, explains the matching principle as:

> The matching principle requires that all revenues earned during a given time period be matched with all the expenses incurred in that same time period to produce these revenues. (at 5)

It follows from this principle that the depreciation of capital assets which have lives that extend beyond one year should be recognized over the useful lives of these assets. As noted by the National Association of Regulatory Utility Commissioners ("NARUC") on page 12 of its *Public Utility Depreciation Practices*, a cost allocation concept for depreciation "satisfies the accounting principle of matching expense and revenues."

From a ratemaking standpoint, depreciation impacts both the revenues from customers, as annual depreciation expense is part of the revenue requirement, and the expenses charged to a given year. If depreciation is such that the depreciation expense over the life of an asset recovers the full cost of the asset (which includes the cost to retire the asset at the end of its life), then

customer rates established based on these depreciation accruals will provide an appropriate return of capital to utility investors who originally provided the capital to invest in the asset.

A related concept in utility ratemaking is "intergenerational equity," in which customers who receive the benefit of a utility asset should pay their fair share of the costs of the asset. If too much depreciation is reflected in the early years of an asset's life, then customers in the early years will pay a disproportionate share of the cost of the asset, while customers in later years will pay less than their fair share. Conversely, if depreciation is too low in the earlier years, then the costs of the asset would be unfairly borne by a future generation of customers.

Q. Does the depreciation of assets that have been recorded in prior years also affect customer rates?

A. Yes. First, whether through the amortization of a reserve variation or through the use of the remaining life technique, depreciation is typically designed to adjust for past levels of depreciation. Thus, if depreciation was too low in the past, current depreciation will typically be

higher, all else equal. There is also an additional way depreciation affects both customer rates and a Company's earnings. Accumulated depreciation, which is effectively a running total of depreciation that has occurred in the past, is a reduction to rate base. As a result, while higher depreciation will have a short-term impact of increasing the revenue requirement in a given year, its longer-term impact will be to reduce rate base and, in turn, the return on rate base. The converse is also true. Lower depreciation expense will have a short-term effect of reducing the revenue requirement but over the long-term will result in a higher rate base and a higher return on rate base. Thus, when considering the impact on customers due to depreciation, there are both short-term and long-term impacts to consider and often these impacts move in the opposite direction from one another.

This concept is explained by NARUC on page 23 of its *Public Utility Depreciation Practices*:

> The regulatory body prescribing depreciation rates is thus confronted with a decision which affects both short-run and long-run interests of the customer and the company. If a commission prescribes rates which yield depreciation accruals that are too low, the revenue requirement in the short run may be lower. But

> the requirements for income taxes and return may offset the apparent savings in depreciation expense, so service rates in the long run may be higher. If depreciation rates are set so low that the revenue requirement fails to repay the capital invested in a group of property by the end of its service life, confiscation takes place or the unpaid cost remains in the rate base until amortized or expensed. On the other hand, if the regulatory body establishes depreciation rates toward the upper end of the zone of reasonableness, rates for service will be higher in the short-run, but may be lower in the long-run.

Q. Does the determination of the appropriate depreciation rates involve the determination of whether a company's capital costs were prudent or reasonable?

A. No. It is important to recognize that depreciation represents the recovery of capital invested by the Company (*i.e.*, the original cost of plant in service less net salvage) in order to provide utility service. The original cost of assets is recorded on the Company's books and recovered through depreciation for assets already in service, a large portion of which have been in service for many years. Further, while net salvage is a future cost that requires estimates of the costs to retire or replace the Company's current asset base, most parties have not challenged net salvage estimates in the depreciation study.

(The one exception is Staff, as we discuss in Section II.B of our rebuttal testimony. For the gas accounts for which Staff has proposed different estimates from ours, they differ by only five percentage points.) This means that, broadly speaking, there is not a major disagreement of the total amount of capital to be recovered through depreciation expense. The question in this case, and the disagreement related to depreciation, is the timing of the recovery of capital – not the amount to be recovered. Because the question is a matter of timing, this means that lower depreciation today means higher depreciation in the future and higher depreciation today means lower depreciation – and a lower rate base – in the future.

It is, therefore, important to consider the implications of the differences of opinion in the timing of recovery. The Company's position is that, due to significant risks of both the obsolescence of the Company's assets and the potential for a smaller customer base to bear the costs of the recovery of capital in the future, the rate at which capital is recovered should be moderately increased in this case as a first step in the process of accounting for this risk. The other parties take the

position that, despite the fact that the Commission and others have recognized some degree of obsolescence is likely to occur and that there is likely to be a smaller customer base in the future, we should defer any recognition of the CLCPA's impacts to a future time. The consequence of these positions would be that, in the future, whatever customer base remains will have to pay for both these deferrals and the return on these deferrals.

Q. Are depreciation estimates based on purely analytical or mathematical computations?

A. No. Even in depreciation studies where there are not considerations similar to the CLCPA, depreciation is a forecast of the future that requires judgement and cannot be based on purely mechanical calculations. For example, NARUC makes clear that there must be a subjective component to estimating service lives. Chapter XIII of *Public Utility Depreciation Practices*, entitled "Actuarial Life Analysis," discusses and emphasizes the subjective nature of the process of estimating service lives. NARUC starts this chapter by explaining that the analysis of historical data is only one part of the process of estimating service lives:

> Actuarial analysis objectively measures how the company has retired its investment. The analyst must then judge whether this historical view depicts the future life of the property in service. The analyst takes into consideration various factors, such as changes in technology, services provided, or capital budgets. (at 111)

Thus, NARUC clarifies that the process of estimating service lives is more than merely an objective measurement of the past. In describing the determination of a survivor curve estimate (referred to as the "projection life" in this passage), NARUC states on page 126 of its *Public Utility Depreciation Practices*:

> The projection life is a projection, or forecast, of the future of the property. Historical indications may be useful in estimating a projection life curve. Certainly the observations based on the property's history are a starting point. Trends in life or retirement dispersion can often be expected to continue. Likewise, unless there is some reason to expect otherwise, stability in life or retirement dispersion can be expected to continue, at least in the near term.
>
> Depreciation analysts should avoid becoming ensnared in the mechanics of the historical life study and relying solely on mathematical solutions. The reason for making an historical life analysis is to develop a sufficient understanding of history in order to evaluate whether it is a reasonable predictor of the future. The importance of being aware of circumstances having direct bearing on the reason for making an historical life analysis cannot be understated. These circumstances, when factored

> into the analysis, determine the application and limitations of an historical life analysis.

Thus, NARUC advises against focusing only on the statistical analysis of historical data, stating that "relying solely on mathematical solutions" should be avoided. Further, we can be quite certain that the future will be different from the past and, therefore, we should expect that service life estimates should differ from the Company's historical experience.

vi. <u>The CLCPA and Obsolescence</u>

Q. Are there differences between the considerations in this case related to the CLCPA and in many other cases in which depreciation is considered?

A. Yes. It is important to recognize that the situation in this case, as it relates to CLCPA, is different from most cases in which the issue of the most reasonable depreciation rates is considered. In other instances, even ones in which obsolescence is an important consideration – for example, the early retirement of coal-fired generation or the replacement of legacy meters with AMI meters – a utility would still be expected to have a consistent or growing customer base. In such circumstances, the issue of

timing, while important from the perspective of intergenerational equity, is not as pronounced as it is in this case. The risk of extending the costs of obsolete assets beyond their economic lives in more common circumstances does not have as disproportionate an impact on the overall customer base. In many more typical instances of obsolescence only a portion of a Company's system is affected (*e.g.*, a power plant or specific asset class) and, as a result, if costs were recovered over an additional five years, for example, it would be likely that most of the customers still paying for assets would be the same as those five years earlier. Thus, in these types of cases of obsolescence, the policy decision before a commission would be to weigh the potential for intergenerational inequity against the potential for rate shock. Similarly, in instances of estimating service lives under these more common instances of obsolescence, the consequence of error (*e.g.*, estimating too long a service life) – while important – is less consequential because any over- or under-recovery could be made up in the next depreciation study with largely the same customer base (and at a minimum with a similar number of customers).

The same is not true with the situation before the Commission in this case. The margin of error is much smaller, and it is even more important to establish depreciation rates that are not too low. As the Commission has recognized, the CLCPA will result in a significant decline in the number of customers receiving gas service. Further, as discussed previously, various parties, from the NY Climate Action Council to the Commission to NYC, have recognized the potential for obsolescence of gas assets and a decline in customers. The prospect of a significant decline in customers – or even a significant decline in usage per customer – means that if the Commission authorizes too low of a recovery of capital (*i.e.*, depreciation that is too low) any costs deferred will likely be borne by a smaller customer base, multiplying the impact on those customers who remain.

In fact, this concept is recognized in the NY Climate Action Council's Scoping Plan:

> All the information before the Climate Action Council (Council) indicates that achievement of the emission limits will entail a substantial reduction of fossil natural gas use and strategic downsizing and decarbonization of the gas system. The integration analysis scenarios and Advisory Panel recommendations also show a greatly

> diminished use of fossil natural gas. To achieve Climate Act targets, the integration analysis scenarios identified the vast majority of current fossil natural gas customers (residential, commercial, and industrial) will transition to electricity by 2050 and identified fossil natural gas use reductions statewide by at least 33% by 2030 and by 57% by 2035. The existing gas system was developed to meet current demand for fossil natural gas and will need to be strategically downsized as this transition proceeds. A well-planned and strategic transition of the gas system will require coordination across multiple sectors. This coordination will include integrated planning with the decarbonization of the power generation sector and buildout of local electric transmission and distribution systems to meet anticipated increases in electricity demand throughout the State. The integrated planning will ensure the transition is equitable for workers and equitable and cost-effective for consumers without compromising reliability, safety, energy affordability, and resiliency. (at 350)

In this context, it is important to understand that the potential harm to customers from higher depreciation today – a higher short-term revenue requirement spread over today's large customer base – is considerably less than the potential harm to customers in the future due to lower depreciation today. This is an even sharper increase in depreciation (because there will be less time to recover costs by 2050), plus a higher rate base, spread over a smaller customer base. The rate impact of lower

depreciation today is likely to result in exponentially higher rate impacts on future customers.

Q. Should depreciation rates incorporate the potential for obsolescence related to the CLCPA?

A. Yes. Depreciation is, by its nature, a forecast of the future. Because depreciation is a prospective exercise of estimating the future useful lives of a company's assets, depreciation must consider how the future could be different from the past. Importantly, any depreciation proposal, even one that assumes the future will be exactly the same as the past, implicitly incorporates an assumption about the future. By failing to recognize the CLCPA's impacts in their depreciation proposals, or by arguing that they should not be included in depreciation until we know these impacts for certain, other parties are effectively making a specific assumption about the future – that the future service lives of the Company's assets will be the same as the Company has experienced in the past.

It follows that the specific depreciation proposals of Staff, UIU, and other parties are based on an assumption that there will be no obsolescence resulting from the CLCPA. In our judgement, this is not a realistic

assumption about the future. Based on likely impacts related to CLCPA, we can be fairly certain that the assumption inherent to their proposals - that the Company's future experience will be the same as its historical experience – is incorrect. Indeed, it is hard to believe, for example, that NYC expects the future to be the same as the past, given that NYC plans to disallow gas service in new construction within a few years. (See The New York City Council, Int 2317-2021, https://nyc.gov.) Accordingly, because the future will be different from the past, we need to incorporate this expectation into estimates of depreciation.

vii. Response to Mr. Garrett's Arguments

Q. Please address Mr. Garrett's conclusion on page 4 of his testimony, that "the Commission must ultimately determine if the Company has met its burden of proof by making a convincing showing that its proposed depreciation rates are not excessive."

A. Mr. Garrett cites (at 6) to passages from *Lindheimer v. Illinois Bell Telephone Co.*, a U.S. Supreme Court case decided in 1934, in support of this statement. We first

note that the specific issues in *Lindheimer* were different from the instant case, as a significant issue in the *Lindheimer* case was the development of the appropriate rate base, to which past depreciation expense was partially relevant. At issue was whether the use of original cost or fair value were appropriate, as well as the calculation of accumulated depreciation. Unlike today, it had not yet become common practice to determine rate base using original cost and the book recorded level of accumulated depreciation. Further, to our knowledge, Illinois did not have a law analogous to the CLCPA that would significantly impact the telephone industry at the time.

Notwithstanding these different circumstances, Mr. Garrett uses a statement from *Lindheimer* that a company has "the burden of making a convincing showing that the amounts it has charged to operating expenses for depreciation <u>have not been excessive</u>" (emphasis added). He then interprets this to mean that the Company must make (at 10) "a convincing showing regarding the CLCPA's specific impacts to service lives." However, there is a significant difference between making a convincing showing that depreciation is not excessive and making a convincing

showing of the precise impacts of the CLCPA over the next 30 years. The latter is, of course, impossible since no one can predict what will occur over the next three decades with that sort of precision. However, there is ample support, such as we provided in Exhibit___(DP-8), that the Company's proposal is not excessive and that it results in much lower depreciation than several potential future scenarios resulting from the CLCPA. We have further demonstrated in Exhibit___(DP-9) that not incorporating CLCPA impacts will disproportionately harm future customers.

Q. Mr. Garrett argues (at 11) that the Company's proposal is "inequitable to current customers." Do you agree?

A. No. We have demonstrated in Exhibit___(DP-9) that our proposal does not disproportionately favor current customers and, to the contrary, Mr. Garrett's proposal is inequitable to future customers. Mr. Garrett's argument appears to be based on the false premise that if service lives are shorter than indicated by historical experience, then the results are necessarily inequitable to current customers. This is incorrect. Intergenerational equity for today's customers will be determined based on the

actual lives the Company's current assets will experience, not what happened in the past. Further, while we do not know yet what those actual lives will be, we can be quite certain that the CLCPA will result in the future being different from the past. Therefore, equity for both current and future customers requires that we account, at least minimally, for the gas transition that has already begun.

Q. Mr. Garrett also claims (at 10) that the Company "makes various conclusory assumptions about the potential impacts of the CLCPA as it relates to service lives but does not provide direct empirical evidence support regarding how the CLCPA will specifically reduce service lives." Do you agree?

A. No. Mr. Garrett's claim ignores a large part of our direct testimony. In our direct testimony and Exhibit___(DP-5), we provided the results of life cycle analysis that demonstrate how the CLCPA will shorten service lives. Indeed, this analysis indicates that service lives should be expected to be shorter than our proposal.

viii. Conclusion

Q. Given all of these considerations, what do you conclude regarding the proposals for gas depreciation?

A. Nothing in the other parties' testimonies provides a reasonable basis to exclude potential impacts of the CLCPA into depreciation. We have provided the results of several modeling scenarios related to the CLCPA, both in our testimony for Con Edison and, for Mr. Allis, for other utilities in New York and other states with similar laws to the CLCPA. We have provided evidence demonstrating the impacts to the Company's service lives which supports our recommendations as a conservative proposal to begin to address these impacts.

The results of all of our modeling illustrate that most scenarios for gas assets result in significantly higher depreciation expense than incorporated into the Company's proposed revenue requirement, as well as higher depreciation expense than recommended in Gannett Fleming's Depreciation Study. Accordingly, the Company's proposal is a minimal recognition of the potential impacts presented in the Depreciation Study.

2. Electric Service Life Estimates

Q. Do you agree with the proposed changes to electric service life estimates made by Staff and UIU?

A. No.

Q. Please address the electric account for which both Staff and UIU have proposed changes.

A. Staff and UIU have both proposed changes for Account 367, Underground Conductors and Devices. We disagree with their proposals and believe their estimates would not recover the costs for these assets in a timely manner. We first note that the service life proposed in the Depreciation Study is the same as the currently approved life for this account. Further, the range of lives typically used by other companies within the industry for these assets is 45 to 55 years. While there are reasons why assets in Con Edison's system in these accounts are different from other utilities, we do not believe underground conductors will last as long as recommended by Staff and UIU. Additionally, there are reasons to expect shorter lives than was the case in the past. We anticipate that assets in the account will be replaced more frequently than they were in the past. The Company expects increases in

electrification of heating and transportation will increase load and potentially require upgrades to existing infrastructure to be able to accommodate these changes.

Our average service life estimate for this account is 55 years. This life is an average and assumes that many of the assets will last well beyond 55 years. This is the same estimate as currently used for this account. It is also the same estimate that Staff recommended in the Company's previous depreciation study three years ago. We are not aware of an evidenced-based reason why Staff's panel would now believe these assets will be in service, on average, 10 years longer than Staff believed three years ago.

Further, one cannot only look at the average service life. Exhibit___(DP-10) provides a graphical comparison of each of the estimates for the account. Our 55-R0.5 estimate incorporates an expectation that only a relatively small subset of assets in the account will last more than 100 years. In contrast, Staff expects almost 20 percent of the account to be in service more than 100 years, which is an expectation for long service lives for a significant number of underground conductors. We do not believe this

is reasonable, nor is UIU's expectation that more than 10 percent of the account will be in service more than 100 years.

Given all of these considerations, the longer lives proposed by Staff and UIU, which are increases from the existing estimate, are less reasonable than what we propose. In our judgement, these expectations inherent to Staff's and UIU's proposals are not reasonable, particularly because of the increased importance of reliability and the need to upgrade assets to accommodate further electrification.

Q. Please address the estimates proposed by UIU for other electric accounts.

A. UIU witness Garrett proposes alternative estimates for two other electric accounts, Account 353, (Transmission) Station Equipment, and Account 362, (Distribution) Station Equipment, that are not realistic judgements for the assets studied. Realistic judgement for these accounts includes consideration of electrification trends, as well as other factors that Mr. Garrett does not appear to have adequately included in his proposals. As discussed in our direct testimony beginning on page 57, load growth and other

factors will mean higher rates of replacement of substation assets than has occurred historically.

For Account 353, the transformation of the transmission grid to incorporate new energy sources, including offshore wind and other renewables, will result in significant changes to the transmission system, including replacements of substation assets and even entire substations. However, Mr. Garrett proposes to increase the average service life for this account by 9 years at a time when we should expect shorter lives. Further, his proposal is primarily the result of his interpretation of older data points from the historical analysis. He incorrectly concludes that these data points indicate that retirements will slow considerably as the assets in the account age. Exhibit___(DP-11) shows a comparison of our proposal (which is the same as Staff's) and that of Mr. Garrett. As can be seen in the graph, both estimates are very similar for the first 40 years of age. Mr. Garrett has selected an unusual L0.5 curve type that forecasts that the rate of retirements declines significantly in later years. This is not a reasonable expectation for this account. Mr. Garrett further expects that many will remain in service well

beyond 100 years with some even surviving to 160 years. We do not believe that these are realistic expectations, especially in the context of climate change impacts (higher temperatures on assets) and increased electrification. Our proposal to continue to use the current service life estimate is, in contrast, more reasonable.

There are similar issues with his proposal for Account 362. We do not believe extending the current average life by five years is consistent with environmental conditions, load growth, or with efforts to operate the system in as reliable a manner as possible.

B. NET SALVAGE

Q. What have the various parties proposed regarding net salvage?

A. Exhibit___(DP-12) compares the net salvage estimates currently in effect, those proposed in our Depreciation Study, and those proposed by Staff. Mr. Garrett did not propose any changes to these estimates. This table also shows the overall and most recent five-year average net salvage ratios based on the statistical analysis of the Company's historical data using the industry standard

method for analyzing net salvage that was used in the Company's Depreciation Study. As the table shows, the historical data indicate net salvage estimates that are, for most accounts, more negative than any party's estimates. Thus, the data supports estimates at least as negative as we proposed in the Company's Depreciation Study.

Q. What are the reasons for the differences in Staff's net salvage estimates?

A. Based on Staff's testimony, their recommendations are based on different interpretations of the trends in the data.

Q. Do you agree with Staff's net salvage proposals?

A. No. As can be seen in Exhibit___(DP-12), for nearly all of the accounts for which we differ from Staff, the overall average net salvage in the data is more negative than both our proposal and Staff's proposal. Thus, in general terms, the data support our estimates, which move the net salvage estimates closer to the overall indications in the historical data. Additionally, we note that the current estimates for many of these accounts are based on a settlement from the Company's previous case, rather than the estimates from the Depreciation Study. For some of

these accounts, the settlement had limited the change in net salvage estimates and in some instances had actually moved the estimates in the opposite direction of the data. In this context, we believe Staff's decision to limit changes results in net salvage estimates that are not sufficiently negative. For example, for Account 367, Underground Conductors and Devices, our estimate of (90) percent, which is the same as what we recommended in the previous study, is less negative than the overall average of (178) percent, and is much less negative than the recent 5-year average of (334) percent. However, in the previous case, the net salvage estimate was limited to (85) percent in the settlement. The overall average at the time of the last study was (167) percent, so the experience since the previous study has increased the overall average to (178). Staff's proposal does not recognize the trend for more negative net salvage for this account.

This is the case for many of the Company's accounts. We believe continuing to limit the change in net salvage to the degree recommended by Staff results in an insufficient level of net salvage when compared to the historical data.

C. RESERVE VARIATIONS

Q. What have the various parties in this case proposed regarding reserve variations?

A. The Company's proposal is to amortize the reserve variation that exceeds 10% of the theoretical reserve over a 20-year period. This approach is consistent with past practices in New York to address reserve variations. Staff, UIU, and NYC have made proposals similar to the Company's. The only difference is that Staff and UIU have calculated different theoretical reserves and reserve variations due to the use of different service lives and, in Staff's case, net salvage estimates. Generally, all parties agree with the same approach to amortize reserve variations that result from their respective proposals.

II. CONCLUSION

Q. Does this conclude your update and rebuttal testimony?

A. Yes, it does.